

Machine Learning - Other supervision strategies

(69152) DRL

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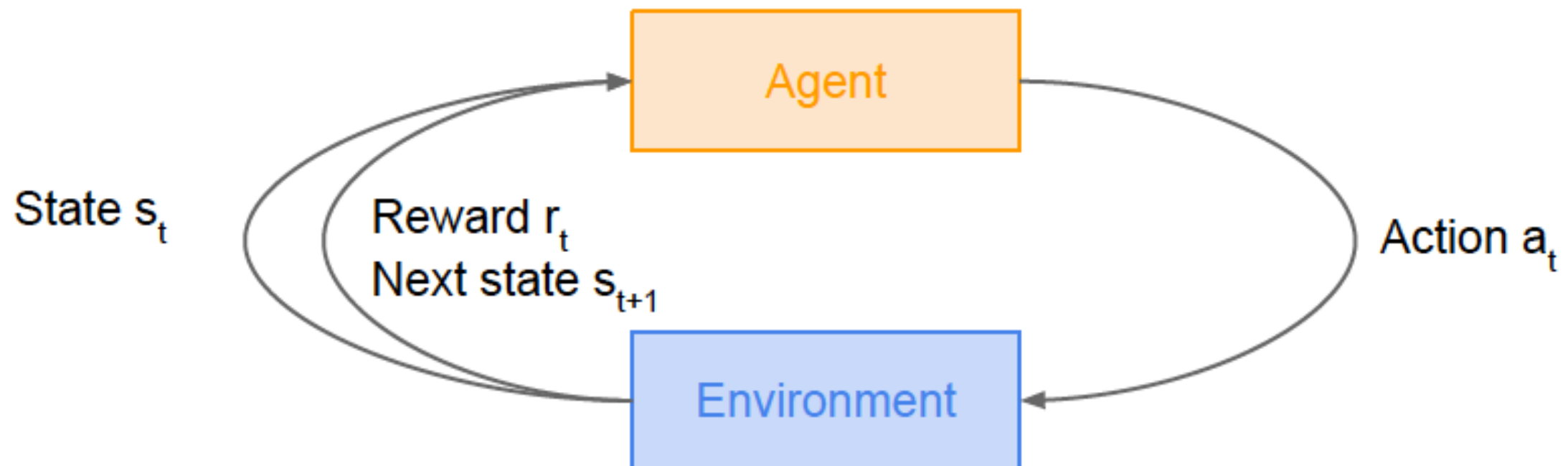
Next ...

- More advanced DL models
- ... and different supervision strategies
 - DRL
 - Unsupervised
 - Recurrent architectures

Deep Learning & RL

- **Reinforcement Learning ingredients:**

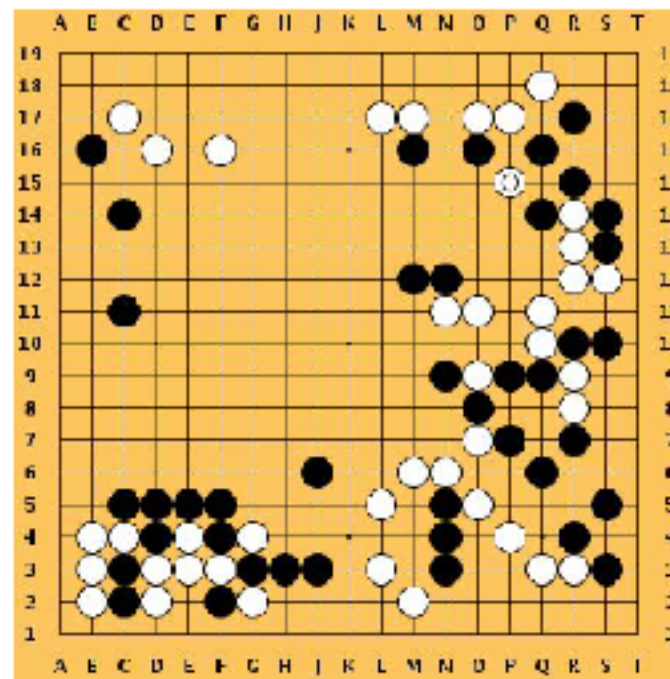
- **Agent** that interacts with the environment
- ... by performing **actions**
- Environment provides numeric **reward**
- We want to *learn which actions to take* to maximize the reward



Deep Learning & RL

- Reinforcement Learning ingredients:

- Objective?
- State?
- Action?
- Reward?



How to model a given problem?



Atari: Mnih, Volodymyr, et al. "Playing atari with deep reinforcement learning." arXiv preprint arXiv:1312.5602 (2013).

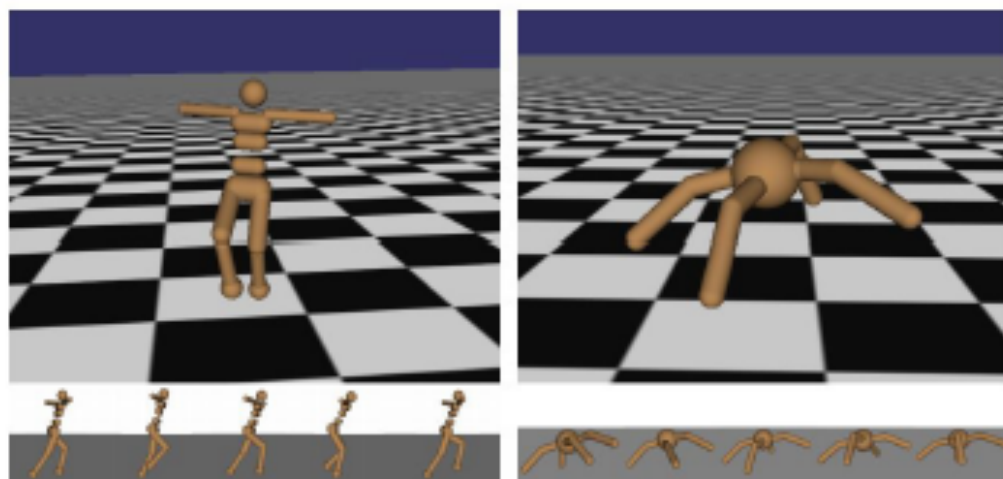
Go: Silver, David, et al. "Mastering the game of Go with deep neural networks and tree search." Nature 2016

Deep Learning & RL

- Reinforcement Learning ingredients:

- Objective
- State
- Action
- Reward

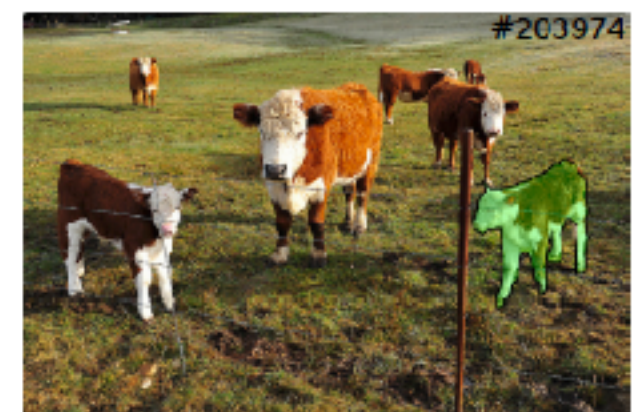
How to model a given problem?



Learning to run & stand up



Is it a person? **No**
Is it an item being worn or held? **Yes**
Is it a snowboard? **Yes**
Is it the red one? **No**
Is it the one being held by the person in blue? **Yes**



Is it a cow? **Yes**
Is it the big cow in the middle? **No**
Is the cow on the left? **No**
On the right? **Yes**
First cow near us? **Yes**

Q&A games

Locomotion: Duan, Yan, et al. "Benchmarking deep reinforcement learning for continuous control." ICML 2016.

GuessWhat: Harm de Vries, et al. GuessWhat?! Visual object discovery through multi-modal dialogue. CVPR 2017.

Deep Learning & RL

Once we get to “represent” the problem ...

How does it work?

Deep Learning & RL

- **Reinforcement Learning:**

- Involves problems about making decisions and/or predictions about the future
 - Examples: *Video games, Board games, Robotics, Recommender systems, ...*
- Our goal is to learn the best *behaviour policy*

Deep Learning & RL

- **RL** —> Markov Decision Process (Markov property*)

- s : state $\in \mathbf{S}$

* *Current state completely characterises the state of the world*

- a : action $\in \mathbf{A}$

- $\mathbf{R}$: reward given a pair (s,a)

- $\mathbb{P}$ probability of transition from s , given a , to next state

- γ : discount factor (future vs present reward)

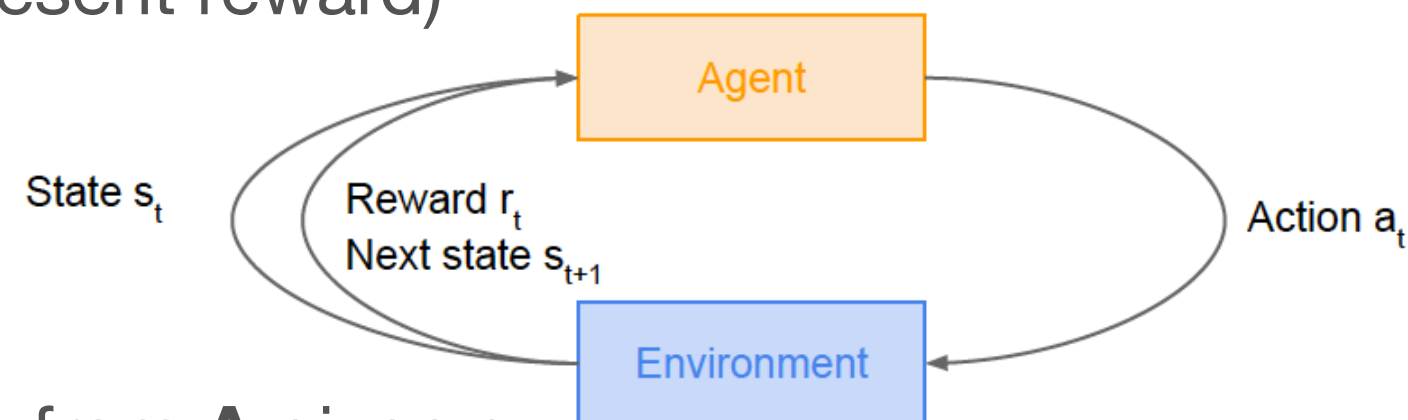
- Policy π :

- function to map from $\mathbf{S}$ to $\mathbf{A}$

- i.e, specify what action to take from $\mathbf{A}$ given s

- **optimal policy** —> **Find it!**

- » maximizes cumulative *discounted* reward $\sum_{t>0} \gamma^t r_t$



Deep Learning & RL

How can we learn the best policy?

Many options ...

- Learn values of each action
- Learn policy directly
- Learn a model - infer policy by planning

Deep Learning & RL

- **Goal: Find optimal policy (π)**
- *How do we use the policy?*
 - Following a policy produces sample paths of actions
 - Each action modifies the state of the environment

Deep Learning & RL

- **Goal:** Find **optimal policy** (π)

How good am I doing?

- **Value function ($V^\pi(s)$)** at state s is the expected cumulative reward after following the policy π from that state s
- **Q-value function ($Q^\pi(s,a)$)** at state s and action a , is the expected cumulative reward performing a in s and then following the policy

$$V^\pi(s) = \mathbb{E} \left[\sum_{t \geq 0} \gamma^t r_t \mid s_0 = s, \pi \right]$$

$$Q^\pi(s, a) = \mathbb{E} \left[\sum_{t \geq 0} \gamma^t r_t \mid s_0 = s, a_0 = a, \pi \right]$$

Deep Learning & RL

- **Goal:** Find **optimal policy** (π)

How good am I doing?

- **Value function** ($V^\pi(s)$) at state s is the expected cumulative reward after following the policy π from that state s
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Deep Q-learning?

Q-learning: Use function approximator to estimate the action-value function:

$$Q^\pi(s,a; \theta) \approx Q^\pi(s,a)$$

function approximator $\rightarrow$ deep neural network $\rightarrow$ deep q-learning

Deep Learning & RL: Deep Q-Learning example

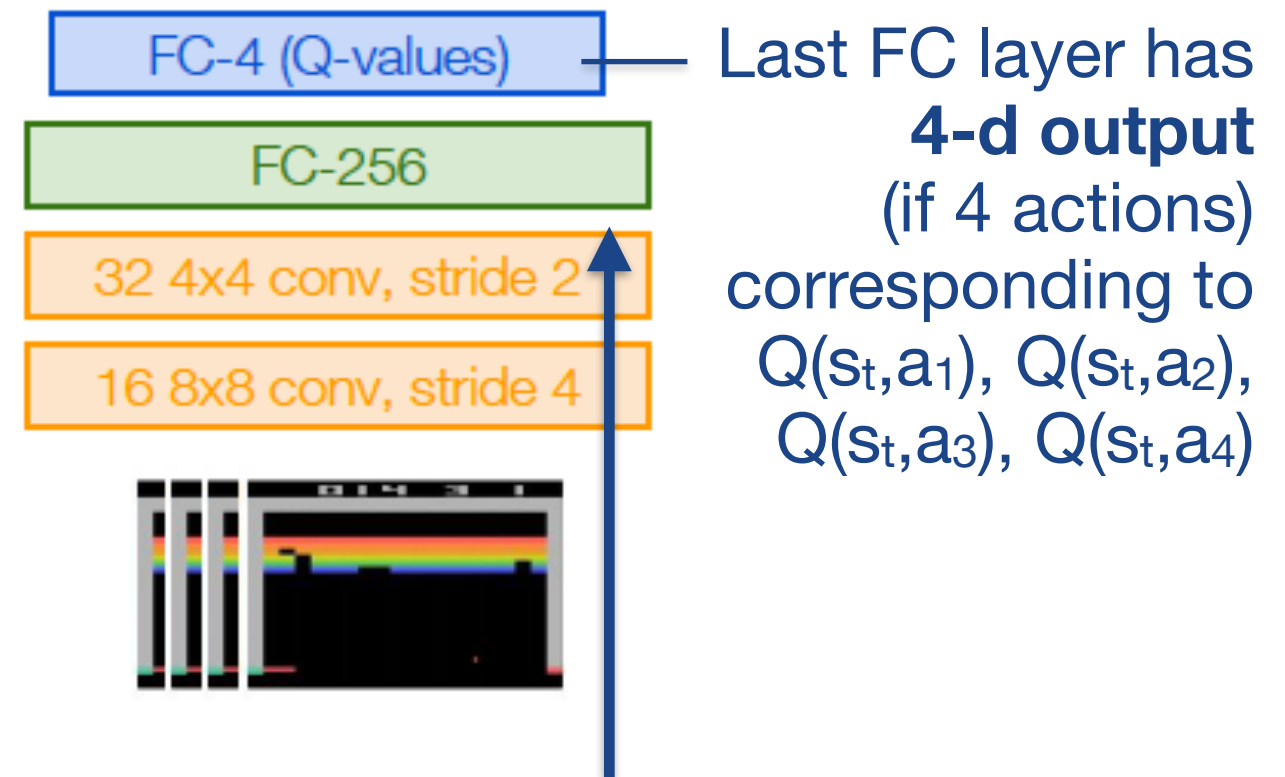
- **Q-value function** $Q^\pi(\mathbf{s}, \mathbf{a})$ at state $\mathbf{s}$ and action $\mathbf{a}$: expected cumulative reward performing $\mathbf{a}$ in $\mathbf{s}$ and then following the policy.

Approximation with $Q^\pi(\mathbf{s}, \mathbf{a}; \theta)$

- **Forward pass?** Q-values for all actions from current state
- **Problems with the consecutive information feed** correlated samples - inefficient learning

Lab 5

$Q^\pi(\mathbf{s}, \mathbf{a}; \theta)$
 θ represents the neural net



Current state $\mathbf{s}_t$: 84x84x4 (last 4 frames after grayscale conv., downsampling and cropping)

Deep Learning & RL

- Sometimes **Q-learning** is not enough ...
 - too complex/too many dimensions,
hard to learn every pair (s,a)
 - Alternative: learn directly the policy —>
policy gradient approaches
- Learn *Policy Gradients* and Q-learning: **actor-critic**
 - training both the **policy (actor)**
and the **Q-function (critic)**,
only on the **pairs (s,a)** generated by the actor

Deep Learning & RL

- Learn Policy Gradients and Q-learning: actor-critic
- Training both the **policy (actor)** and the **Q-function (critic)**, only on the pairs (s,a) generated by the actor

*A well known
Deep RL example*



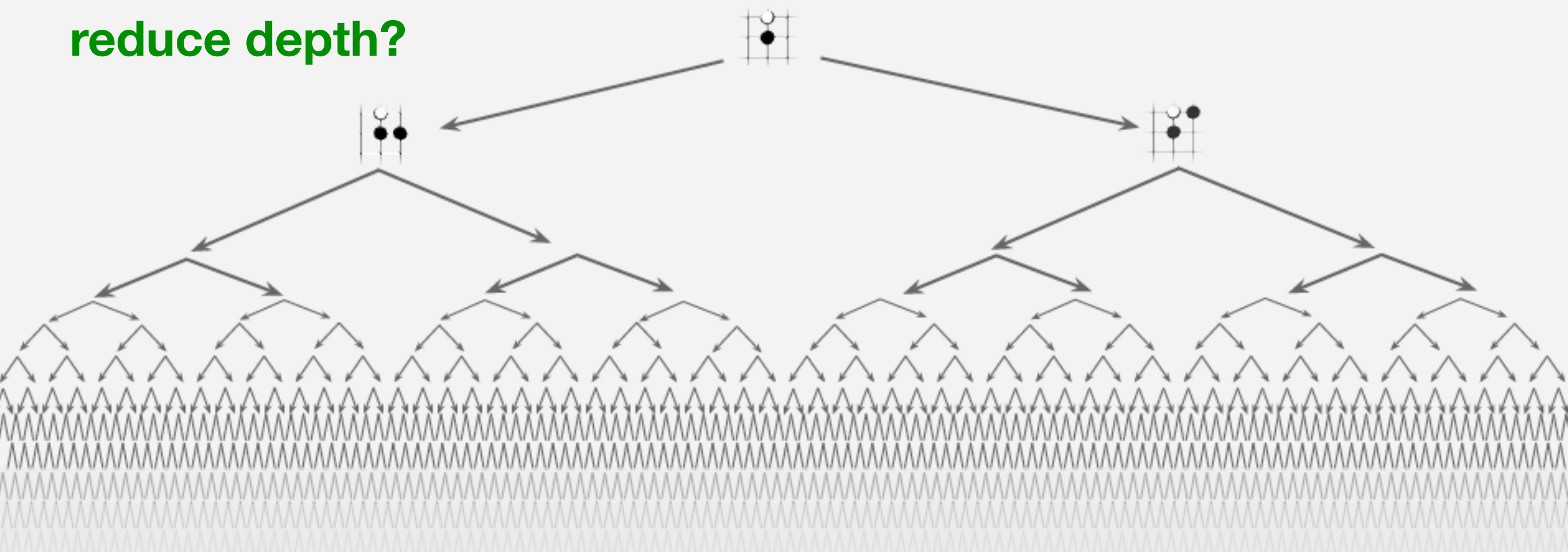
Deep Learning & RL

- supervised + reinforcement
- tree search + deep RL



Exhaustive search

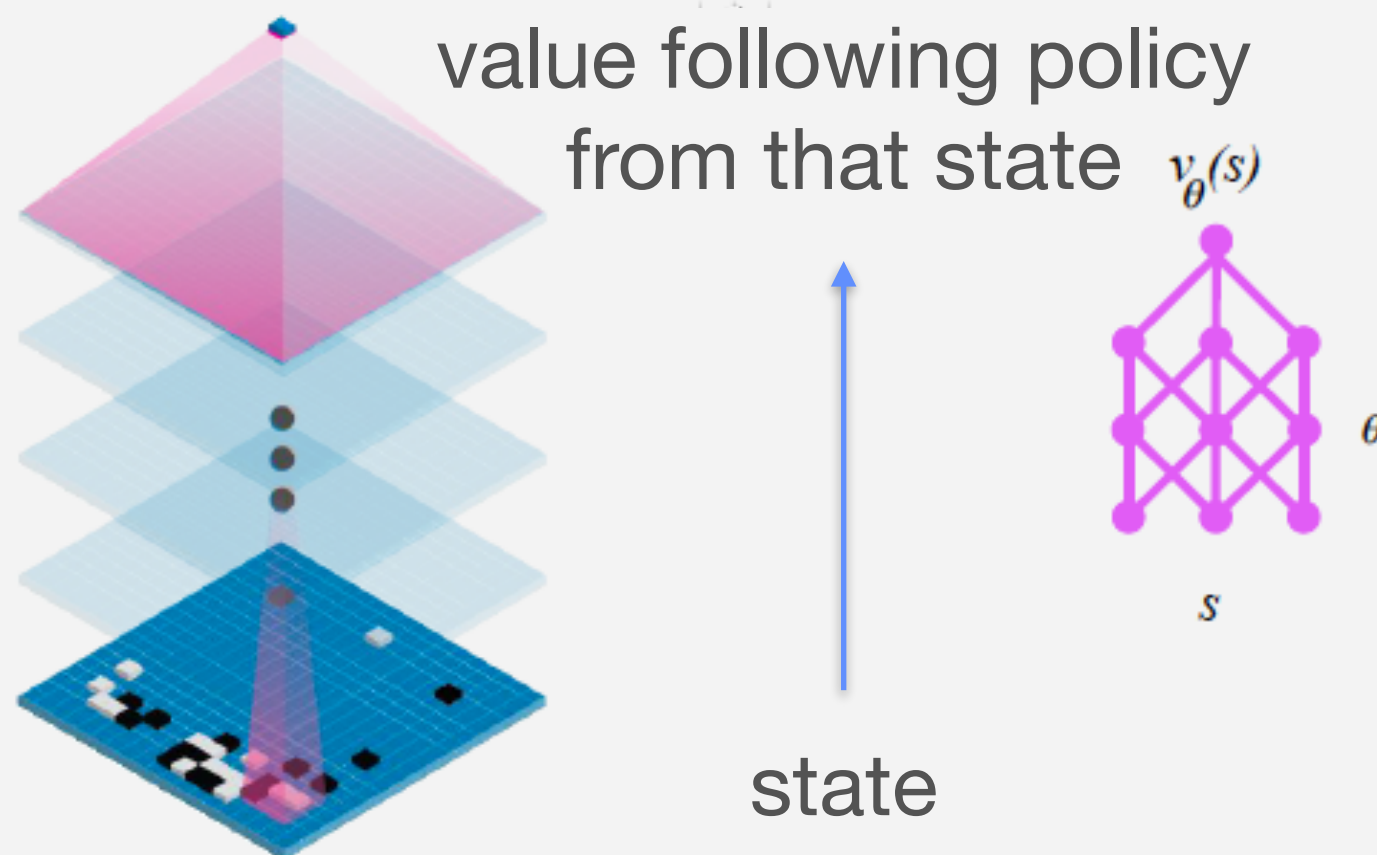
reduce depth?



Deep Learning & RL

- supervised + reinforcement
- tree **search** + **deep RL**
 - both the policy (*actor*) and the **Q-function** (*critic*)

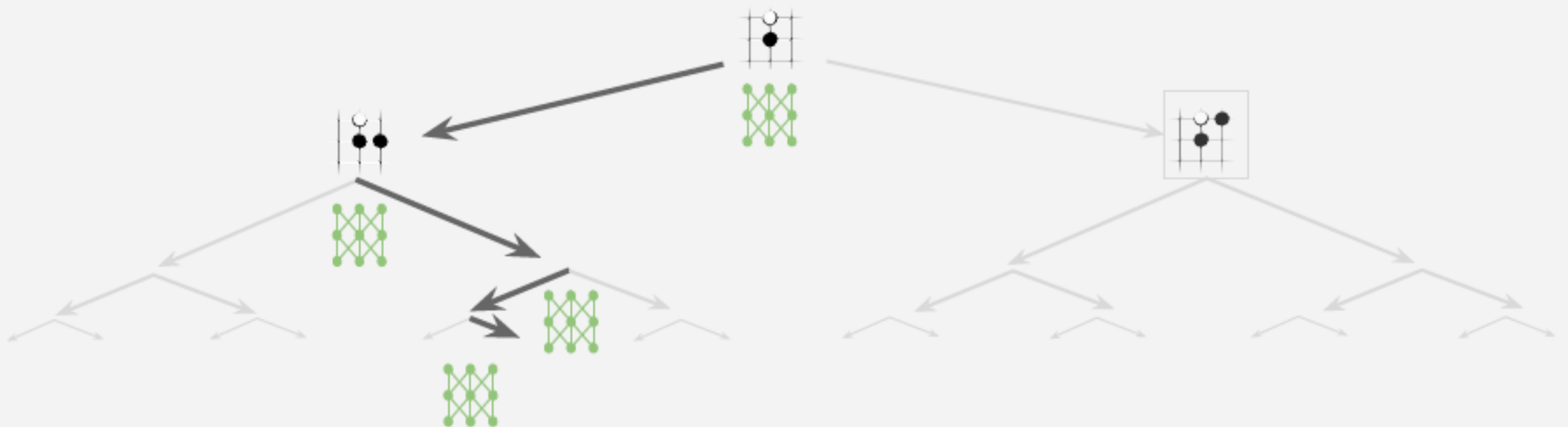
Reducing depth with value network



Deep Learning & RL

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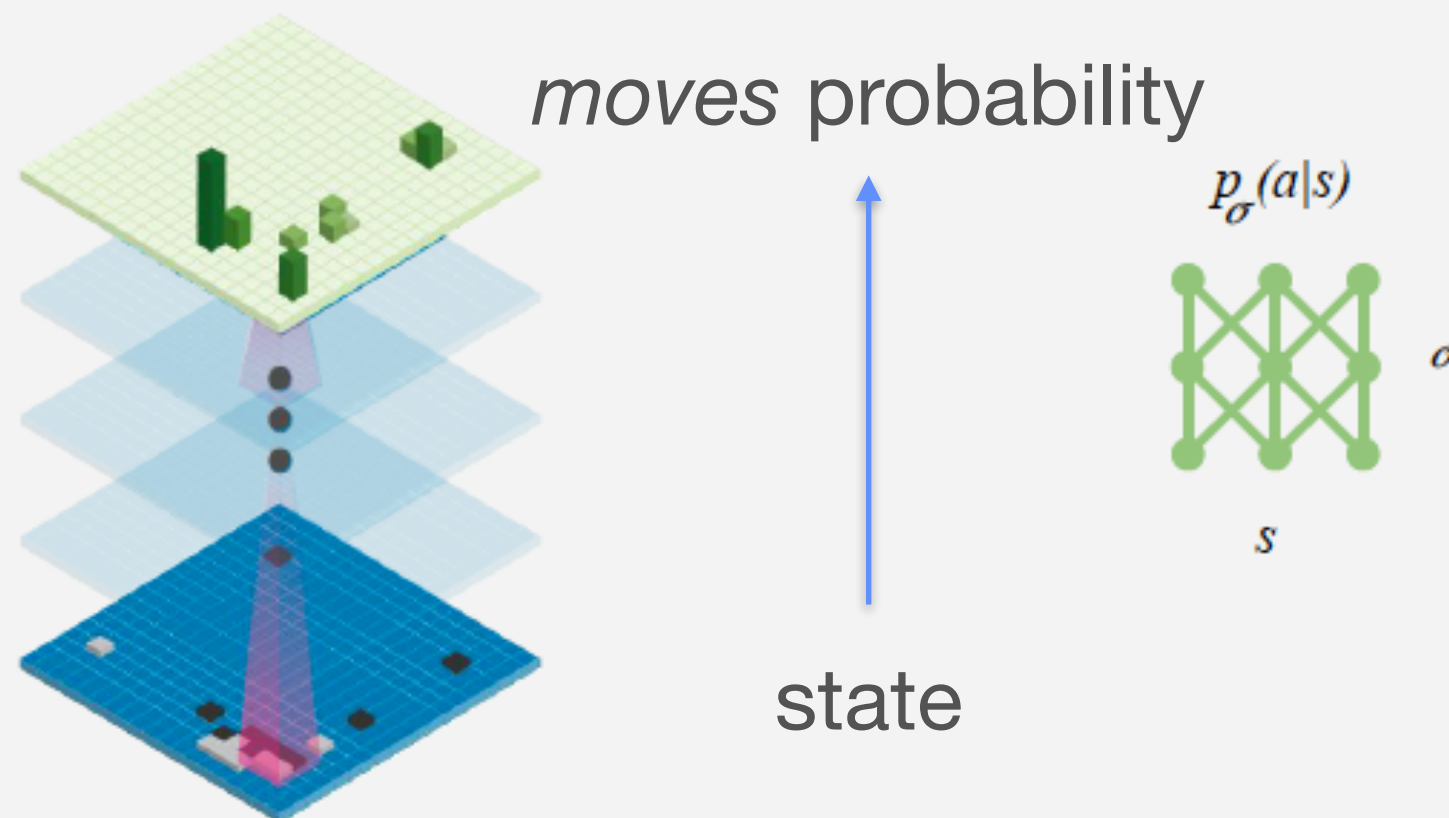
Reducing breadth with policy network



Deep Learning & RL

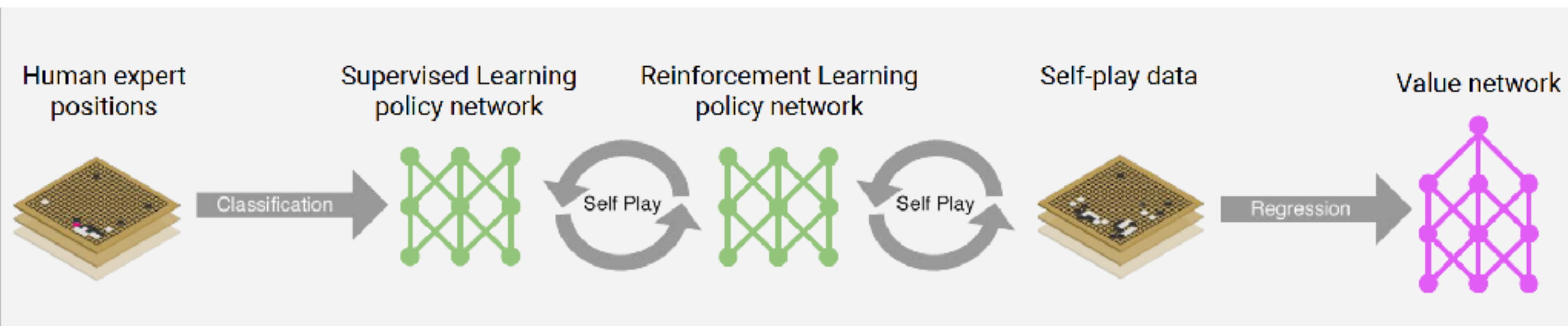
- supervised + reinforcement
- tree **search** + **deep RL**
 - both the **policy** (*actor*) and the Q-function (*critic*)

Reducing breadth with policy network



Deep Learning & RL

- supervised + reinforcement
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Deep Learning & RL

Another **Deep RL** example: learning 2 players

Oracle & *Guesser* networks

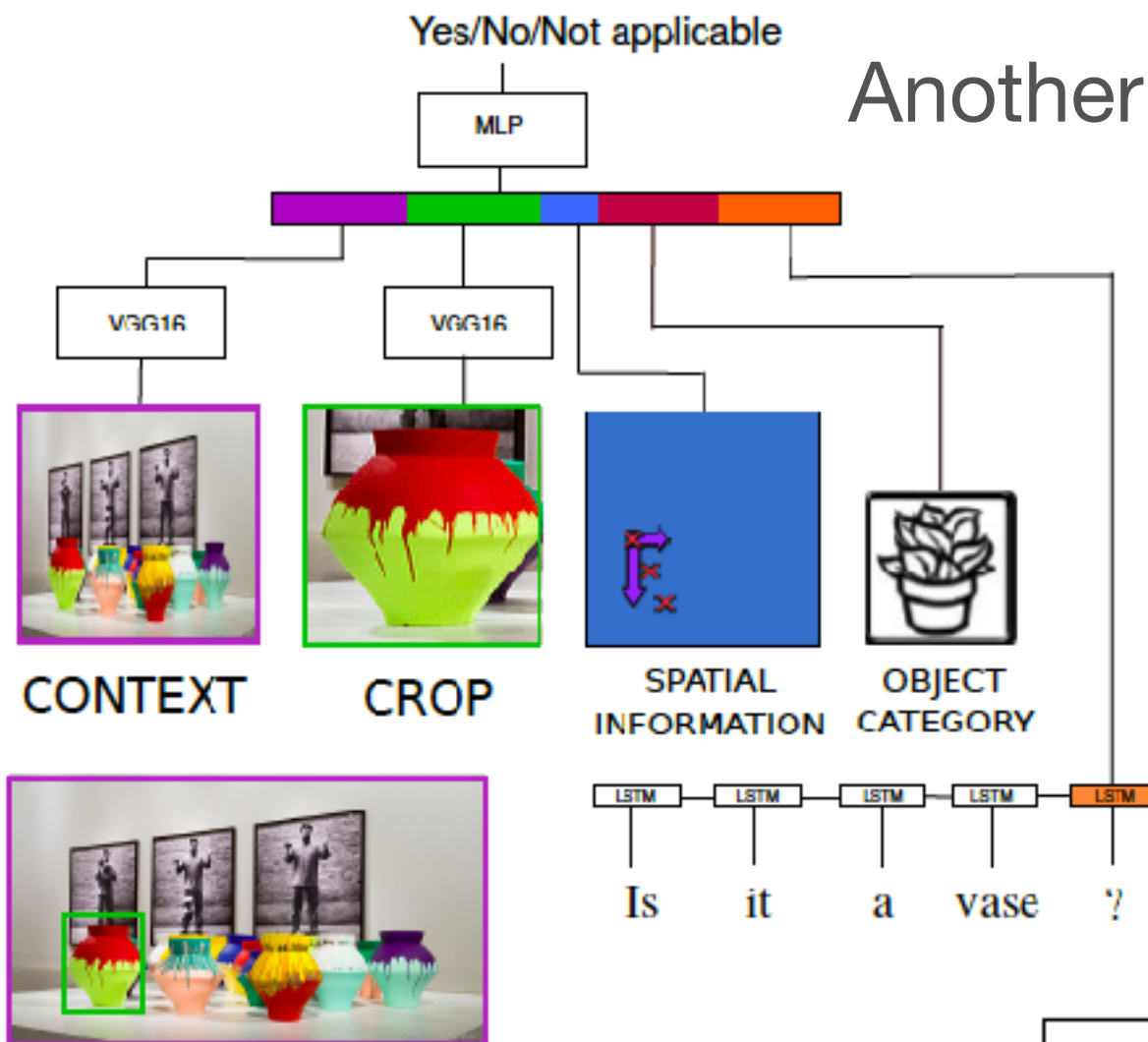
guesser:

- model of image with segmented objects (all MLPs share weights)
- generates questions

oracle:

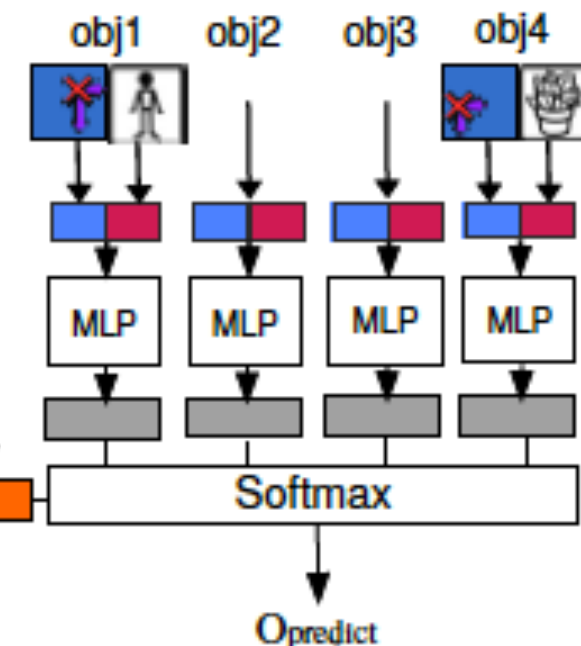
- model of image, question, crop, spatial info and category
- *knows the answers*

Harm de Vries, et al. GuessWhat?! Visual object discovery through multi-modal dialogue. CVPR 2017.



Is it a vase? Yes
Is it partially visible? No
Is it in the left corner? No
Is it the turquoise and purple one? Yes

LSTM / HRED encoder



Deep Learning & RL

- Still a lot of on-going research
 - Policy gradients: general - high variance - requires a lot of samples.
 - Q-learning: does not always work (if it does, pretty efficient)

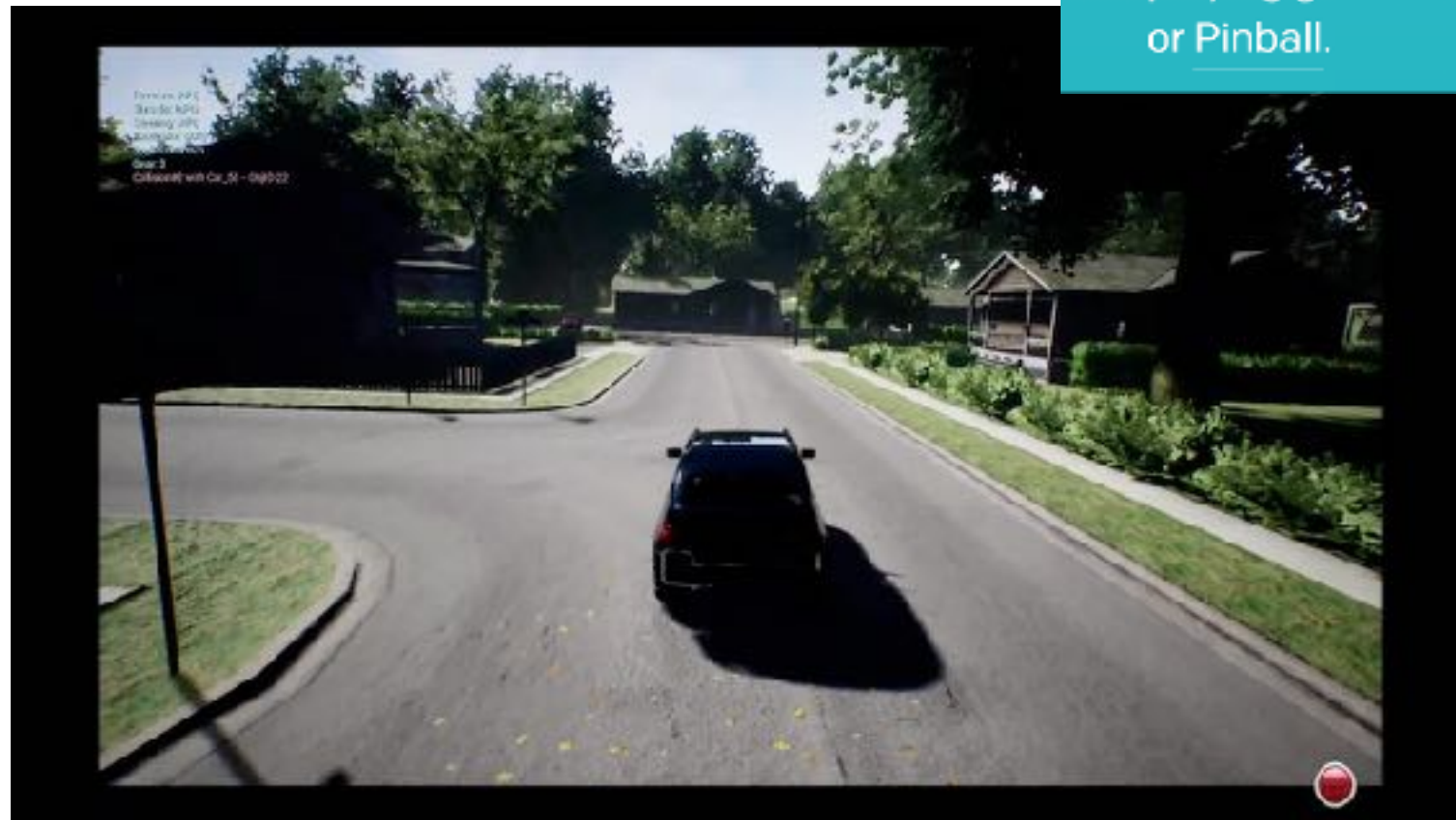
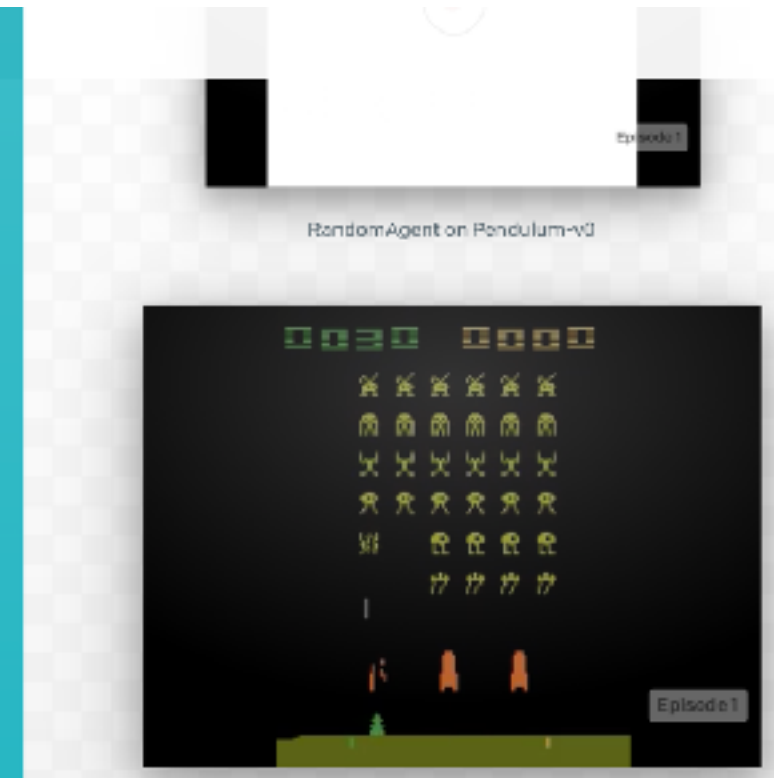
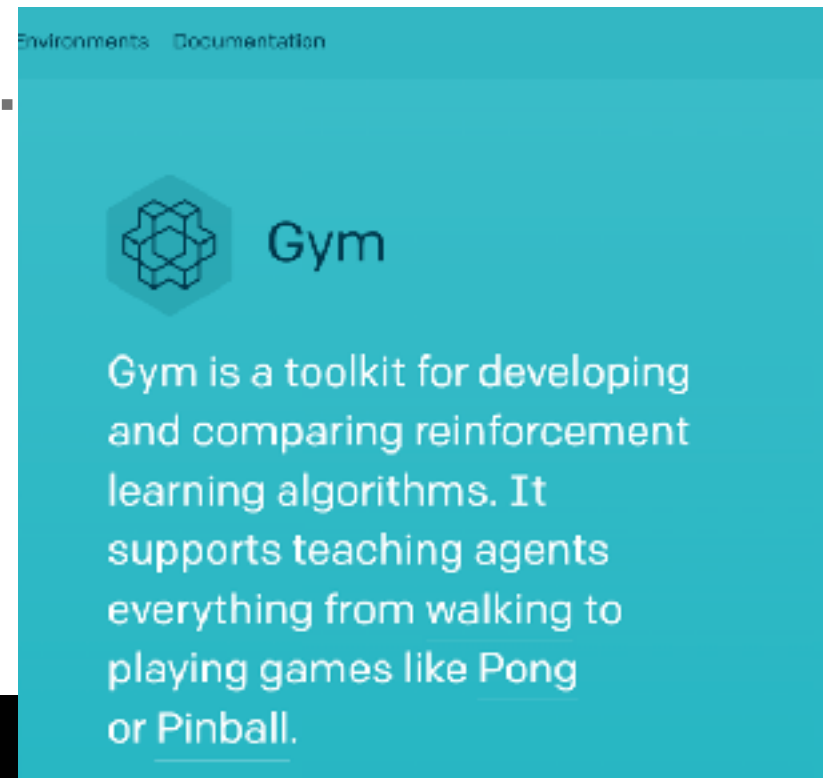
Some of the challenges?

Deep Learning & RL

- Still a lot of on-going research
 - Policy gradients: general - high variance - requires a lot of samples. *How do we sample?*
 - Q-learning: does not always work (if it does, pretty efficient). *How do we explore all the pairs s,a ?*

Deep Learning & RL: simulation frameworks

- An essential piece ...



[OpenAI GYM](#)

[Microsoft AirSIM](#)

[SONY GT](#)

Next ...

- Other supervision strategies:
 - Unsupervised
 - GANs



Bibliography - Resources for some of the materials today

- Stanford classes on deep learning for Computer Vision (<http://cs231n.stanford.edu>) and Deep Learning (<https://cs230.stanford.edu/>)
- Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016. <http://www.deeplearningbook.org>
- Deep Learning Summer School Montreal: <https://mila.quebec/en/cours/deep-learning-summer-school-2017/>
- [CS 294: Deep Reinforcement Learning](#) and [CS294-129 Designing, Visualizing and Understanding Deep Neural Networks](#). UC Berkeley.
- DRL Bootcamp: <https://sites.google.com/view/deep-rl-bootcamp/lectures>
- [Open AI Gym](#)